

Improving Automated Scoring of Prosody Using Deep Learning Algorithm

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Abstract

This study proposed and evaluated an improved approach for automated scoring for prosody in oral reading fluency assessment by focusing on specific prosodic features and including the samples from text-to-speech (TTS) audio in the training. This research explored different models by using deep learning with different prosodic and frequency-related acoustic features to produce better performance, especially in cross-domain validation, where phrases are assumed to be unknown. The results demonstrated that the model with selected prosodic features had better cross-domain performance with an accuracy of 59.4% compared to 50.4% of the previous research. Our finding also demonstrated that the students have more weight to influence the estimating performance of cross-domain validation compared to the passages.

Keywords: Automated Scoring, Oral Reading Fluency, Deep Learning

Word count: [2,787 words.]

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Introduction

Oral reading fluency (ORF) is the ability to read a text quickly, accurately, and with proper expression (National Reading Panel, 2000). Prosody is one dimension of ORF that is the ability to read smoothly with an expression that presents the meaning of the text by different stresses, pitch variations, intonations, rates, phrasings, and meaningful pausing (Rasinski, 2004).

Traditionally, human raters use rubrics to score the prosodic features of ORF. Some attempts to automatically score prosody have been made in the field. Recently, Sammit et al. (2022) incorporated selected frequency-related voicing features, such as pitch and fundamental frequency, achieving a correctness accuracy of 86.4%. However, it was for in-domain classifications, where phrases were assumed to be known. When their trained algorithm was applied to cross-domain contexts, where phrases were assumed to be unknown, their accuracy rate dropped to 52.6% and 60.4%. Therefore, this current study attempts to improve the accuracy rate for automated scoring for prosody, specifically for cross-domain contexts.

The use of specific prosodic features has been demonstrated satisfactory in the literature (Eyben et al., 2016; Schuller et al., 2010; Weninger et al., 2013). Thus, we focused on how we can combine selected specific prosodic and spectral features with the deep learning neural network structure to improve cross-domain performance. Also, we evaluated the effect of including features extracted from text-to-speech (TTS) audio to the deep learning neural network

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structure to improve cross-domain accuracy as Sammit et al. suggested for a possible future study.

Literature Review and Related Work

A computer-based assessment system of ORF will significantly reduce the assessment administration effort, and some attempts to automatically score prosody have been made in this domain. As an earlier attempt, Duong and others incorporated prosodic contours by focusing on prosodic features, such as latency, duration, mean pitch, and mean intensity (Duong et al., 2011; Mostow & Duong, 2009). Bolaños et al (2011, 2013) have investigated an automatic approach for assessing children's oral reading prosody using lexical and prosodic features to train a machine learning classifier to classify the expression of students' oral reading based on the NAEP (National Assessment of Education Progress) four levels scale (accuracy of 90.93% classifying two categories "fluent" or "non-fluent" and overall 76.05% for marking 1-4 levels). Sabu and Rao (2018) provided an automated assessment for ESL children's ORF by using speech recognition and prosodic modeling with precision-recall scores of 73.2% and 73% for phrase boundary detection and 59.2% and 80% for prominence detection. The latest explored research that focused on automated prosody classification with a machine learning approach achieved so far, the highest performance for in-domain (known phrases are used in the training set) prosody classification of 86.4% (Sammit et al., 2022). However, the classification accuracy dropped to 50.4% and 51.0% when applied to cross-domain (unknown phrases) with the NAEP four levels.

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We suspected that the gap between in-domain and cross-domain performance resulted from overfitting of the model since they had resampling to try to balance the samples of classes.

Acoustic features (such as energy-related, spectral, and voicing-related) have been extensively used for studying in the speaker verification field (Ananthakrishnan & Narayanan, 2009; Dehak et al., 2007; Ferrer et al., 2010; Kockmann et al., 2010, 2011; Martínez et al., 2012, 2013). Researchers also found that prosodic features (e.g., loudness, F0 smoothing, zero-crossing rate, etc.) showed higher correlations to emotion recognition for sound, speech, and music (Eyben et al., 2010, 2016; Schuller et al., 2010; Weninger et al., 2013).

Purpose of Research

Similar to some studies, our research is based on a supervised model with four-level rubrics, however, our model focused on using specific features for automated assessment prosody of ORF and comparing the performance of cross-domain among different feature sets. The purpose of this work is to enhance the generalization of the model with prosodic features and to improve cross-domain estimating performance.

Data

The data used in this study is audio-recorded reading data collected by Nese and Kamata (2014-2018) and augmented in Sammit et al. (2022), which includes a total of 5,841 recordings (4,128 (70.7%) in the training/validation sample and 1,713 (29.3%) in the testing cross-domain sample) on 30 reading passages (approximately 50-100 words in length) by 1,335 2nd through 4th

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grade students in the U.S. The demographic information shows in Figure 1. The dataset is imbalanced among four classes. The most of students in each grade are White. Also, English was the first language for a majority of students.

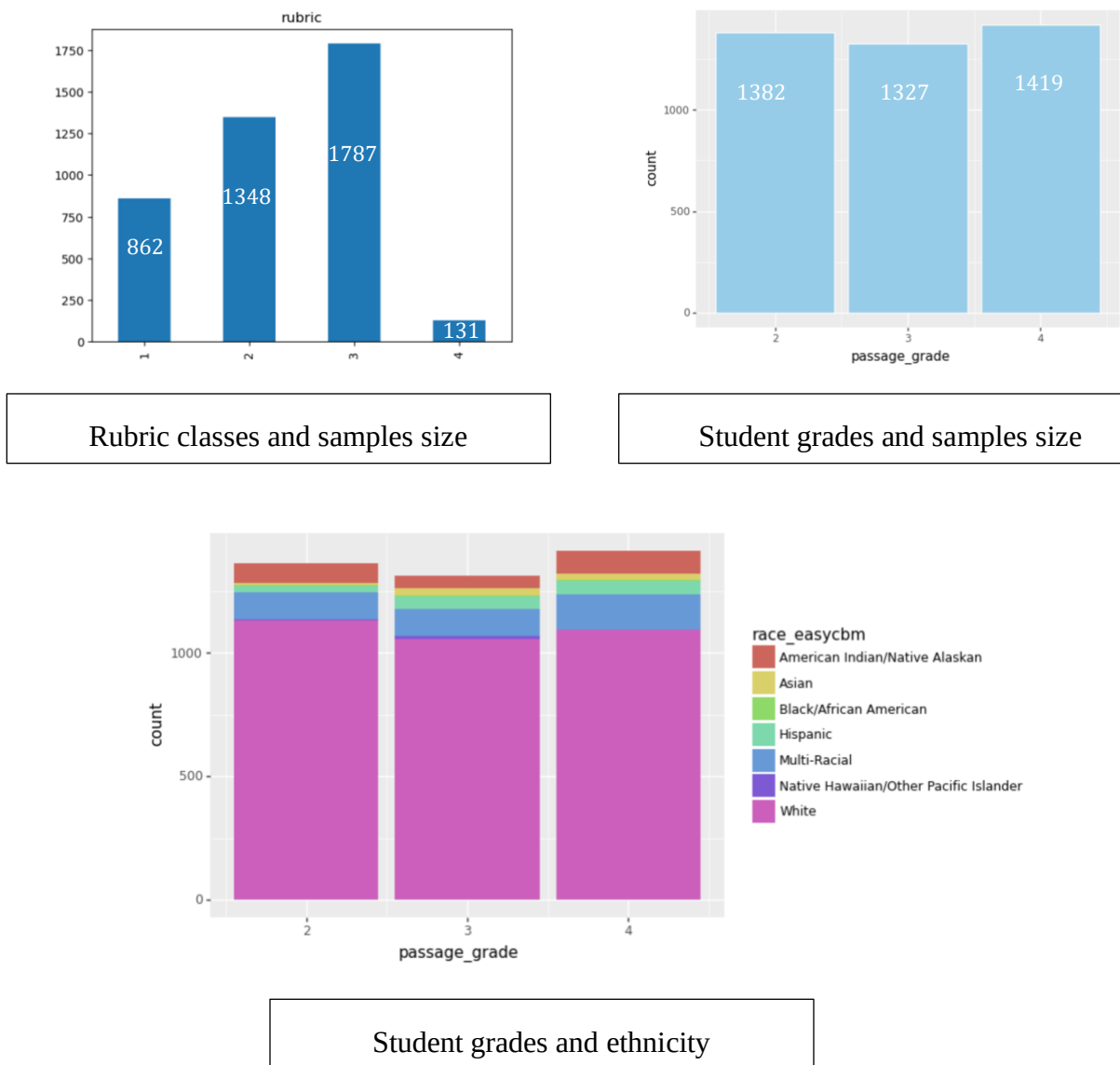


Figure 1. Data demographic information.

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Methods

We used specific prosodic features extracted with an open-source tool, called *openSMILE* in *ComArE_2016* and *eGeMAPs v02* format (Eyben et al., 2010, 2016; Schuller et al., 2016). We focused on their low-level descriptors (LLD), which include 65 features under the *ComArE_2016* set (4 energy-related features, 55 spectral related features, and 6 voicing-related features), and 25 features under the *eGeMAPs v02* set (3 energy-related features, 8 frequency related features, and 14 spectral related features).

Based on the literature review, nine specific prosodic features were chosen under *ComArE_2016* LLD and shown in Table 1 along with their feature names and feature groups.

Table 1. *Selected prosodic features.*

Feature name	Description	Group
F0final_sma SHS	F_0 (SHS and viterbi smoothing)	Prosodic
audspec_lengthL1norm_sma	sum of auditory spectrum (loudness)	Prosodic
audspecRasta_lengthL1norm_sma	sum of RASTA-style filtered auditory spectrum	Prosodic
pcm_RMSenergy_sma	RMS energy	Prosodic
pcm_zcr_sma	zero-crossing rate	Prosodic
pcm_fftMag_fband250-650_sma	spectral energy 250-650Hz	Spectral
pcm_fftMag_fband1000-4000_sma	spectral energy 1k-4kHz	Spectral
pcm_fftMag_spectralCentroid_sma	spectral centroid	Spectral
pcm_fftMag_psySharpness_sma	psychoacoustic sharpness	Spectral

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These types of features have been reported to be relevant for speech emotions in the literature. Nine specific features were chosen as a small set of prosodic features, including 5 prosodic features (e.g., energy features that represent smoothness and loudness, and voicing features that represent fundamental frequency and pitch), and 4 spectral features (e.g., spectral energy at two different frequency ranges, and psychoacoustic sharpness of acoustic signals).

To establish our automated scoring algorithm, we first utilized the model structures used by Sammit et al. (2022), such as X-Vectors (Snyder et al., 2018) or self-attention (Okabe et al., 2018), using categorical cross entropy (CCE) or quadratic weighted kappa (QWK) loss function (de la Torre et al., 2018). Second, we incorporated standard reading audio features from text-to-speech (TTS) into the train data set, which included 300 readings (30 passages each with 10 narrators). The performance of the models was compared among the models that used 5 different features groups: (1) all 65 features, (2) selected 56 features, (3) 9 selected features, (4) 7 frequency-feature in *eGeMAPs* v02 set, and (5) 16 selected features (9 selected features + 7 frequency-feature in *eGeMAPs* v02 set) to evaluate the efficacy of the smaller selected set of features. For each of the five features groups, we ran the models that were with X-Vectors or self-attention model structure, using CCE or QWK loss function, and with or without TTS samples. As a result, a total of 40 models (5 features groups $\times$ 2 model structures $\times$ 2 loss function types $\times$ 2 TTS conditions) were trained for evaluating performance.

Classification agreements with human raters were evaluated by QWK as an indicator of the performance of the models. As the dataset is imbalanced in classes, we used metrics such as

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Micro Accuracy, Micro-avg OvR ROC AUC Score, and Macro-avg OvR ROC AUC Score for measuring the model performance of multi-class classification.

In terms of cross-domain samples, we separated samples into three different groups based on whether or not either or both passages and students appeared in the training process (see Table 2).

Table 2. *Cross-domain groups*

Samples	Description
538	Passages appeared in the training, but students did not.
956	Students appeared in the training, but passages did not.
219	Both passages and students did not appear in the training

Results

We compared all 40 models based on the metrics with testing results of all 1,713 samples as well as each cross-domain group (see Table 3 and Table 4). Overall test result for all 1,713 cross-domain samples showed that the model with TTS samples and selected 56 features of *ComArE_2016* set had the best performance that 57.3% Micro-accuracy, 0.34 QWK, 0.82 Micro-avg OvR ROC AUC, and 0.74 Macro-avg OvR ROC AUC. On the other hand, the model without TTS samples and the same structure had 59.0%, 0.35, 0.82, and 0.65 for each metrics.

Table 3. Comparing the performance of models with TTS samples.

Models with TTS samples				Overall Test 1,713 samples				538 samples (Passage appeared in training)				956 samples (Passage did not in training)				219 samples (Passage and student both did not in training)			
Features set	Classifier	Kappa Loss	Features	Micro Accuracy	QWK	Micro-avg OvR ROC AUC Score	Macro-avg OvR ROC AUC Score	Micro Accuracy	QWK	Micro-avg OvR ROC AUC Score	Macro-avg OvR ROC AUC Score	Micro Accuracy	QWK	Micro-avg OvR ROC AUC Score	Macro-avg OvR ROC AUC Score	Micro Accuracy	QWK	Micro-avg OvR ROC AUC Score	Macro-avg OvR ROC AUC Score
CompArE2016	SelfAttention	0	65	46.4%	0.39	0.75	0.65	45.7%	0.37	0.74	0.65	46.7%	0.39	0.75	0.64	46.6%	0.44	0.76	0.67
	SelfAttention	1	65	55.0%	0.33	0.81	0.67	53.0%	0.31	0.81	0.66	56.1%	0.34	0.82	0.68	55.3%	0.33	0.79	0.65
	SelfWX	0	65	42.1%	0.35	0.68	0.64	44.2%	0.36	0.68	0.66	42.1%	0.38	0.69	0.64	37.4%	0.15	0.66	0.59
	SelfWX	1	65	42.1%	0.16	0.75	0.69	44.2%	0.20	0.76	0.70	42.1%	0.15	0.75	0.69	37.4%	0.07	0.73	0.67
CompArE2016	SelfAttention	0	56	43.8%	0.35	0.71	0.63	42.6%	0.31	0.70	0.64	44.2%	0.37	0.71	0.63	44.7%	0.38	0.70	0.61
	SelfAttention	1	56	57.3%	0.34	0.82	0.74	55.4%	0.33	0.81	0.76	58.7%	0.34	0.83	0.73	55.7%	0.31	0.81	0.75
	SelfWX	0	56	44.1%	0.29	0.73	0.62	41.8%	0.29	0.73	0.64	44.8%	0.29	0.73	0.60	46.6%	0.28	0.73	0.65
	SelfWX	1	56	44.1%	0.15	0.80	0.71	41.8%	0.12	0.78	0.70	44.8%	0.15	0.80	0.71	46.6%	0.18	0.79	0.70
CompArE2016	SelfAttention	0	9	46.0%	0.32	0.73	0.60	45.5%	0.32	0.71	0.56	45.8%	0.33	0.74	0.61	47.9%	0.26	0.72	0.59
	SelfAttention	1	9	56.0%	0.30	0.80	0.67	49.6%	0.25	0.76	0.61	59.4%	0.34	0.82	0.70	56.6%	0.29	0.79	0.68
	SelfWX	0	9	44.4%	0.29	0.73	0.62	42.9%	0.24	0.72	0.64	45.9%	0.32	0.74	0.62	41.6%	0.29	0.69	0.57
	SelfWX	1	9	44.4%	0.14	0.70	0.59	42.9%	0.14	0.70	0.55	45.9%	0.16	0.70	0.61	41.6%	0.09	0.69	0.59
eGeMAPS	SelfAttention	0	7	47.5%	0.35	0.73	0.63	44.1%	0.30	0.72	0.61	48.7%	0.38	0.73	0.63	50.2%	0.34	0.74	0.62
	SelfAttention	1	7	45.4%	0.18	0.71	0.63	46.1%	0.17	0.71	0.63	44.5%	0.18	0.71	0.63	47.9%	0.23	0.71	0.66
	SelfWX	0	7	46.2%	0.31	0.72	0.60	44.8%	0.28	0.71	0.62	47.3%	0.33	0.73	0.59	44.7%	0.27	0.73	0.63
	SelfWX	1	7	46.2%	0.17	0.73	0.60	44.8%	0.17	0.71	0.57	47.3%	0.18	0.74	0.63	44.7%	0.13	0.75	0.58
CompArE2016 _9 + eGeMAPs_7	SelfAttention	0	16	47.6%	0.38	0.74	0.64	44.6%	0.33	0.73	0.64	49.3%	0.40	0.75	0.63	47.9%	0.42	0.73	0.64
	SelfAttention	1	16	53.9%	0.29	0.80	0.74	51.9%	0.27	0.79	0.75	54.9%	0.29	0.81	0.75	54.3%	0.30	0.80	0.73
	SelfWX	0	16	53.2%	0.37	0.79	0.63	46.3%	0.28	0.76	0.59	55.8%	0.39	0.81	0.65	58.9%	0.44	0.81	0.66
	SelfWX	1	16	53.2%	0.23	0.78	0.71	46.3%	0.14	0.78	0.70	55.8%	0.26	0.79	0.71	58.9%	0.31	0.76	0.70

Note: Kappa Loss: 1 means kappa loss; 0 means CCE loss

Table 4. Comparing the performance of models without TTS samples.

Models without TTS samples				Overall Test 1,713 samples				538 samples (Passage appeared in training)				956 samples (Passage did not in training)				219 samples (Passage and student both did not in training)			
Features set	Classifier	Kappa Loss	Features	Micro Accuracy	QWK	Micro-avg OvR ROC AUC Score	Macro-avg OvR ROC AUC Score	Micro Accuracy	QWK	Micro-avg OvR ROC AUC Score	Macro-avg OvR ROC AUC Score	Micro Accuracy	QWK	Micro-avg OvR ROC AUC Score	Macro-avg OvR ROC AUC Score	Micro Accuracy	QWK	Micro-avg OvR ROC AUC Score	Macro-avg OvR ROC AUC Score
CompArE2016	SelfAttention	0	65	47.3%	0.33	0.74	0.63	45.0%	0.28	0.73	0.62	49.0%	0.36	0.75	0.65	45.7%	0.27	0.71	0.56
	SelfAttention	1	65	55.5%	0.31	0.79	0.63	51.1%	0.25	0.77	0.59	58.2%	0.35	0.80	0.65	54.3%	0.29	0.78	0.61
	SelfWX	0	65	49.4%	0.32	0.75	0.57	47.6%	0.30	0.74	0.56	50.1%	0.34	0.75	0.59	50.7%	0.26	0.74	0.54
	SelfWX	1	65	49.4%	0.19	0.83	0.65	47.6%	0.19	0.83	0.66	50.1%	0.19	0.84	0.65	50.7%	0.19	0.82	0.61
CompArE2016	SelfAttention	0	56	47.6%	0.31	0.70	0.60	45.5%	0.33	0.83	0.66	49.6%	0.32	0.71	0.62	43.8%	0.17	0.67	0.56
	SelfAttention	1	56	59.0%	0.35	0.82	0.65	57.6%	0.34	0.81	0.62	59.8%	0.36	0.83	0.66	58.4%	0.34	0.82	0.64
	SelfWX	0	56	47.0%	0.27	0.67	0.58	44.2%	0.18	0.66	0.57	48.1%	0.30	0.68	0.59	48.9%	0.34	0.64	0.54
	SelfWX	1	56	47.0%	0.17	0.66	0.57	44.2%	0.13	0.69	0.58	48.1%	0.18	0.65	0.58	48.9%	0.19	0.61	0.54
CompArE2016	SelfAttention	0	9	48.8%	0.36	0.75	0.59	43.9%	0.29	0.73	0.59	51.9%	0.38	0.77	0.59	47.5%	0.38	0.73	0.57
	SelfAttention	1	9	55.0%	0.30	0.80	0.61	53.2%	0.29	0.78	0.58	57.3%	0.32	0.81	0.63	49.8%	0.21	0.78	0.57
	SelfWX	0	9	52.0%	0.41	0.77	0.60	48.1%	0.37	0.76	0.60	54.5%	0.42	0.78	0.62	50.2%	0.43	0.75	0.55
	SelfWX	1	9	52.0%	0.24	0.81	0.68	48.1%	0.21	0.80	0.65	54.5%	0.26	0.82	0.70	50.2%	0.20	0.80	0.65
eGeMAPs	SelfAttention	0	7	48.4%	0.36	0.73	0.56	46.1%	0.34	0.71	0.54	49.2%	0.35	0.74	0.57	50.7%	0.40	0.73	0.55
	SelfAttention	1	7	50.5%	0.26	0.78	0.64	50.9%	0.26	0.78	0.62	52.0%	0.28	0.79	0.67	42.9%	0.15	0.74	0.58
	SelfWX	0	7	45.9%	0.29	0.73	0.63	45.7%	0.27	0.72	0.61	45.2%	0.28	0.72	0.62	49.3%	0.34	0.75	0.67
	SelfWX	1	7	45.9%	0.17	0.76	0.65	45.7%	0.18	0.75	0.63	45.2%	0.15	0.77	0.65	49.3%	0.21	0.76	0.66
CompArE2016 _9 + eGeMAPs_7	SelfAttention	0	16	47.0%	0.30	0.74	0.61	43.5%	0.23	0.71	0.56	49.1%	0.32	0.75	0.62	46.6%	0.33	0.73	0.62
	SelfAttention	1	16	56.3%	0.32	0.83	0.68	53.2%	0.29	0.81	0.65	58.3%	0.34	0.84	0.70	55.3%	0.29	0.82	0.63
	SelfWX	0	16	48.1%	0.30	0.74	0.60	43.7%	0.27	0.71	0.56	48.8%	0.31	0.75	0.60	55.7%	0.36	0.76	0.65
	SelfWX	1	16	48.1%	0.20	0.78	0.65	43.7%	0.15	0.77	0.63	48.8%	0.20	0.79	0.67	55.7%	0.31	0.77	0.61

Note: Kappa Loss: 1 means kappa loss; 0 means CCE loss

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The model with TTS samples and selected 9 prosodic features of *ComArE_2016* with the cross-domain group that passages were not but the students were in the training process, showed the best performance with 59.4% Micro-accuracy, 0.34 QWK, 0.82 Micro-avg OvR ROC AUC, and 0.70 Macro-avg OvR ROC AUC. Even though this performance was not better than the model without TTS samples with 56 features that has the performance as 59.8%, 0.36, 0.83, 0.66, respectively, given that the model only with 9 prosodic features, the result implies a potential impact of prosodic features for improving estimating performance. Through all models and cross-domain groups, the model with selected 9 prosodic features of *ComArE_2016* + 7 frequency features of *eGeMAPs* showed the highest QWK score 0.44.

Discussion and Future Work

The implications of our main findings are described as follows. First, the train data set was imbalanced with very small samples of class 4 compared to class 1 to 3. Therefore, it might have influenced the overall classification performance measured by QWK. We also tested adding 300 standard TTS audios (scoring as class 4) to train samples. However, compared to the models without standard TTS audios data, the result showed that they did not help improve the model performance. The reason might be that the sample size was still very small. Another possible reason is that TTS audios were adult narrators compared to the test samples that all are 2nd to 4th-grade students. Second, when we considered the cross-domain test of ORF, there were two aspects of it, namely passage and student. The passage or student or both might have been unknown for a model. Initially, we suspected that passages would have more weight than students

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in model performance for score classifications. However, our results revealed that the samples with students known in the training process showed better-estimating performance than the samples with students unknown in the training process in all models. Third, the two models with selected 9 prosodic features, testing with samples that both passage and student unknown in the training process showed overall the best performance of QWK compared to the model with more features. This result may imply that the model with specific prosodic features had a higher generalization to capture the common characteristics of prosody. This study can be an important demonstration for achieving a satisfactorily high correctness accuracy and classification agreement rate with a set of fewer selected features, which would be more efficient and interpretable.

We noted some limitations to this work. First, the train samples included readings from 1,335 students. Although it was seemingly a large sample for a deep learning neural network, it might still have not been large enough to have the neural network learn scoring prosody with desirable accuracy. Second, most of our sample students are White students with English as their first language. Therefore, the model has limited generalizability when used for different racial groups of students as well as for students whose first language is not English. Additional investigations are needed to find out whether the model performance depends on the classes, cross-domain groups, and grade levels. Also, the impact of specific types of features on classification performance needs to be investigated.

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